Z1 UNION-ANCHOR ANALYSIS

Evidence Record: Shared DFT Anchors

A projected saving, and what happened when we ran it. A frozen, artifact-backed analysis of 29 analyzable diffusion paths compares four separately evaluated sparse-anchor sets with their shared union.

EVIDENCE STATUS	FIGURES	REFERENCES
Projected accounting, no new DFT. Internally reviewed (AI-assisted); not peer-reviewed.	**1 selected**	**3 cited**

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AUTHOR / INSTITUTION
Alex Welcing · Lupine Science

CANONICAL SOURCE
lupine.science

01 / EXECUTIVE SUMMARY

What the evidence supports

Four model guides nominate sparse images on the same diffusion paths. Deduplicating identical within-path image indices reduces the locked projected reference-evaluation count from 558 to 154. The baseline is four separate runs.

- **Finding.** 72.4% fewer DFT evaluations, projected. The frozen separate-model accounting contains 558 projected DFT evaluations; the shared union contains 154. The baseline is four separate runs.
- **Ceiling.** The paths have 5–7 images, 167 in all, and the union covers 154 of them; deduplication alone caps the saving at 73.1%.
- **Exclusion.** One panel path lacks a converged prediction profile in all four artifacts and is excluded rather than imputed.
- **Boundary.** The result is prediction-side accounting, not an executed-DFT, accuracy, cost, or universal-transfer claim.
- **Realized pilot.** In July 2026 the preregistered 40 meV gate against published VASP barriers failed (path 7: 118.8 meV). A same-engine check adopted after that failure passed (0.0–6.8 meV) with every image evaluated: Open (same-engine only). Error against the published references: 693.0–705.7 meV.

EVIDENCE VERDICT

Supported as projected accounting only; not an accuracy result.

The committed JSON record, input SHA-256 digests, analysis script, and separately locked figure support the 558-to-154 comparison on 29 analyzable paths. Missing model–path profiles remain explicit and no realized campaign denominator is substituted. The realized pilot did not show accuracy against published references.

Reader orientation

This record presents the locked figure and table first, then the selection rule, exclusions, published literature, and audit trail. The full account of the pilot is “What happened when we ran it” on the canonical article.

02 / KEY FIGURES

The claim, then the mechanism

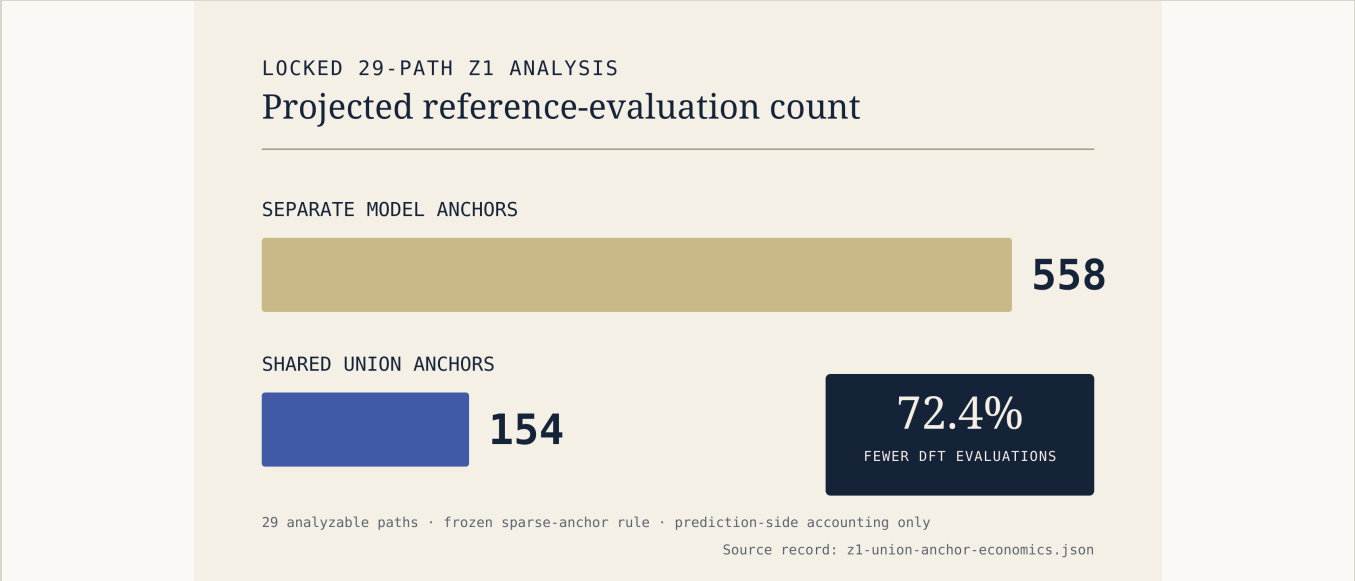


Figure 1. Frozen evaluation-count comparison on the 29 analyzable Z1 paths. The bars report the separate-model anchor count and the deduplicated within-path union count.

Source: Article source notes

03 / DATA AND METHOD

Inspectable values and assumptions

Table 1. Projected counts under the frozen sparse-anchor rule; no DFT was run for this analysis.

ACCOUNTING BASIS	ANALYZABLE PATHS	PROJECTED DFT EVALUATIONS
Four model-specific anchor sets, counted separately	29	558
Within-path union of identical image indices	29	154

METHOD RECORD

UNIT OF ANALYSIS

Model–path sparse-anchor set and within-path union of image indices

EVIDENCE CUTOFF

2026-07-21

VERSIONING

LS-PP-2026-006 version 1.1.0, 2026-11-12 (version 1.0.0: 2026-08-03). Figure input and generated PDF checksums are in the evidence-record manifests.

KNOWN LIMITATIONS

One locked panel, four recorded universal interatomic-potential guides, one sparse-anchor rule, prediction-side accounting only. No accuracy, realized execution, dollar, wall-time, energy, emissions, or transfer claim is made.

Methodology note

Recorded predicted image-energy profiles select sparse anchor indices under a frozen rule. The analysis compares the sum of available model-specific sets with the cardinality of their within-path union and runs no new DFT.

Builds run entirely from repository assets. No runtime network requests, remote fonts, analytics, animation, or hidden interactive state are permitted in the PDF path.

AUTHOR AND INSTITUTION CREDITS

Author
Alex Welcing

Institution
Lupine Science

Editorial review
Internally reviewed (AI-assisted) and approved for publication 2026-08-03; not peer-reviewed

Data and figure provenance
Locked Z1 panel, recorded four-model campaign artifacts, committed analysis JSON and script, citation identifier audit, and SHA-256 manifests

04 / BIBLIOGRAPHY

Sources and audit trail

1. A climbing image nudged elastic band method for finding saddle points and minimum energy paths Journal of Chemical Physics, 2000. https://doi.org/10.1063/1.1329672
2. Low-Scaling Algorithm for Nudged Elastic Band Calculations Using a Surrogate Machine Learning Model Physical Review Letters, 2019. https://doi.org/10.1103/PhysRevLett.122.156001
3. CHGNet as a pretrained universal neural network potential for charge-informed atomistic modelling Nature Machine Intelligence, 2023. https://doi.org/10.1038/s42256-023-00716-3

Audit links

Canonical article, including “What happened when we ran it”: https://lupine.science/articles/shared-dft-anchors/

Released data and reproduction pack: https://lupine.science/data/savings-stack-v1/README.md

Citation identifier audit: https://lupine.science/data/savings-stack-v1/citation-verification-2026-07-21.md